Milaan

Closing the books before noon

Payment reconciliation for the analyst who has four files and one close to finish.

The Singleton · Hemanth Chilakala

ExaThon India 2026 · Track 1

Meet the analyst

> A reconciliation analyst at a mid-sized Indian NBFC can't close the day's books on time because UPI, NACH and payment-gateway settlement files arrive in different formats with mismatched references, so they chase breaks by hand in Excel. If this worked, they'd close the books before noon and spend their time only on the exceptions that actually matter.

Milaan reconciles synthetic receipts for the fictional Sahyog Finance Ltd,
using Exasol Personal for normalisation, matching, scoring and audit records.

One credit. 37 payments.

37 PayFast payments

Gross ₹10,02,336.29

− fees ₹8,018.71 − GST ₹1,443.38

Bank credit ₹9,92,874.20

36 ledger receipts present · 1 missing

Loan SFL-BL-3979018 · missing receipt ₹31,698.77

Why spreadsheets fail

References disagree
Truncated bank narrations and different receipt identifiers.

Amounts disagree
Gross EMIs, net settlements, fees and GST.

Dates disagree
Delayed credits, holidays and next-day NACH returns.

Illustrative consequences: suspense balances, wrong customer follow-ups and slower audit review.

Milaan in one picture

Ledger · Bank · Gateway · NACH

↓ bulk IMPORT

EXASOL PERSONAL

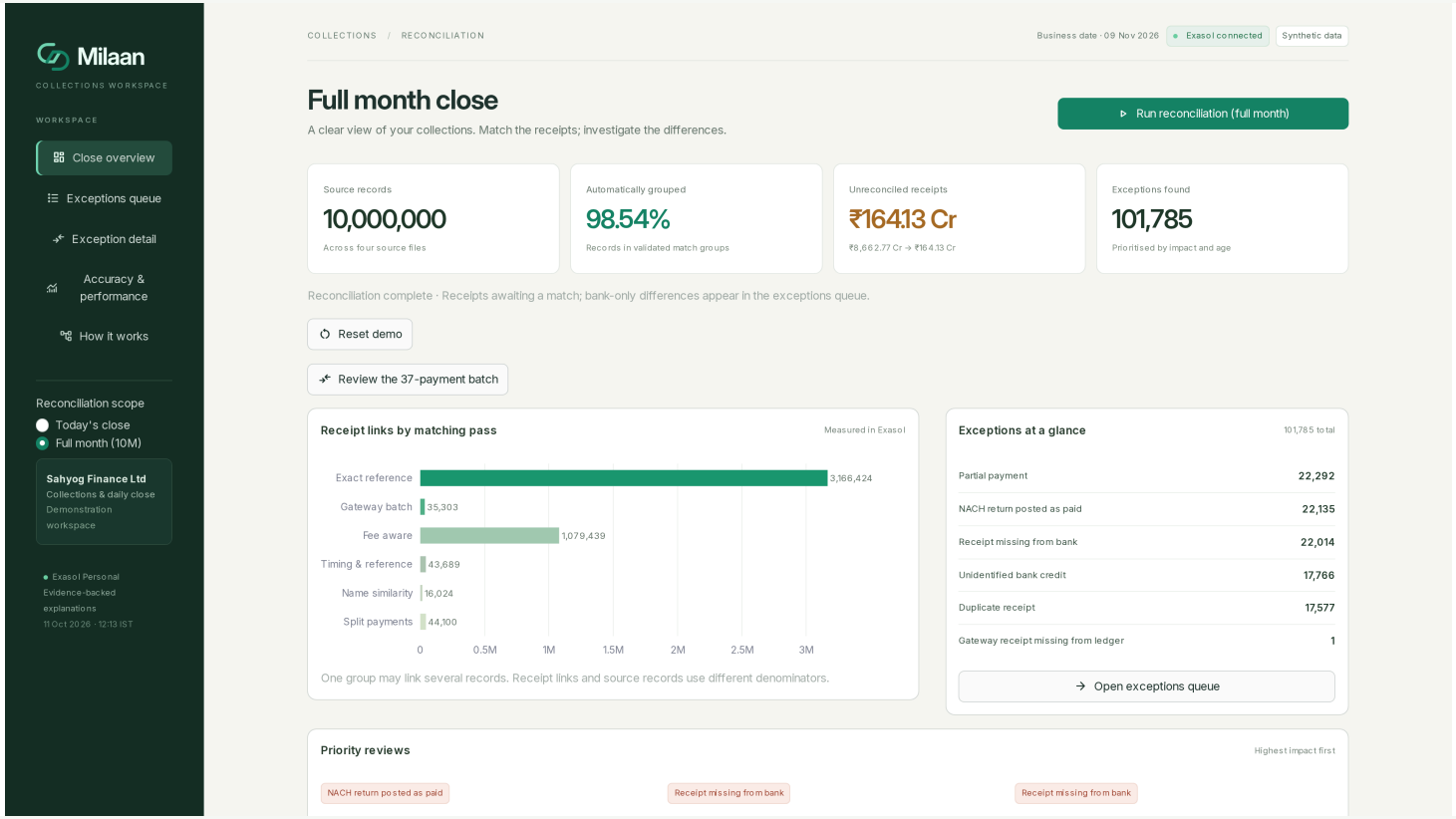
SQL normalisation → seven passes + UDFs → ranked exceptions

↓ small result sets

Analyst review → decision audit in Exasol

Separate ground truth → SQL scoring

Watch the close progress



14 Oct 2026 · 29.49s core · 9.97s scoring · 39.46s total

Full month for scale · Today's close for arriving records and carried open items

Seven passes. One shrinking queue.

EXACT	**3,166,424**	3,166,424 cumulative
PG_BATCH	**35,303**	3,201,727 cumulative
FEE_AWARE	**1,079,439**	4,281,166 cumulative
TOLERANT	**43,689**	4,324,855 cumulative
FUZZY	**16,024**	4,340,879 cumulative
SPLIT_MERGE	**44,100**	4,384,979 cumulative
CLASSIFY	**0**	4,384,979 cumulative

Measured 10M run · counts are receipt links · CLASSIFY labels remaining exceptions

Why Exasol

88.23 seconds

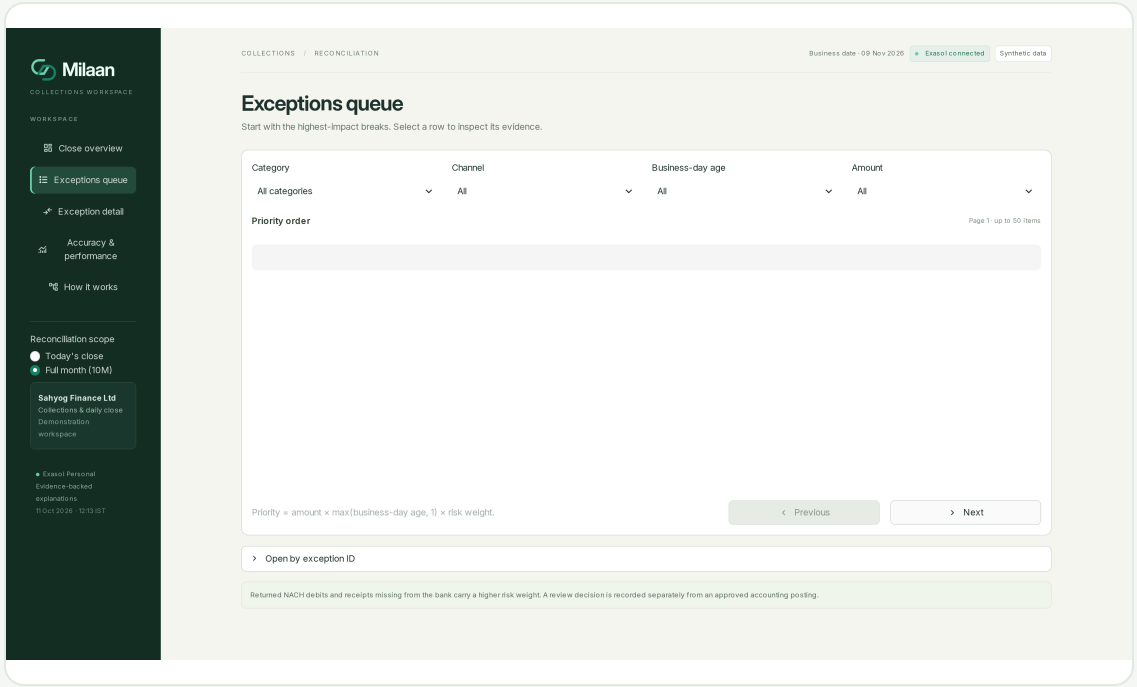
10,000,000 source records · 4 cores · 15.62 GiB RAM

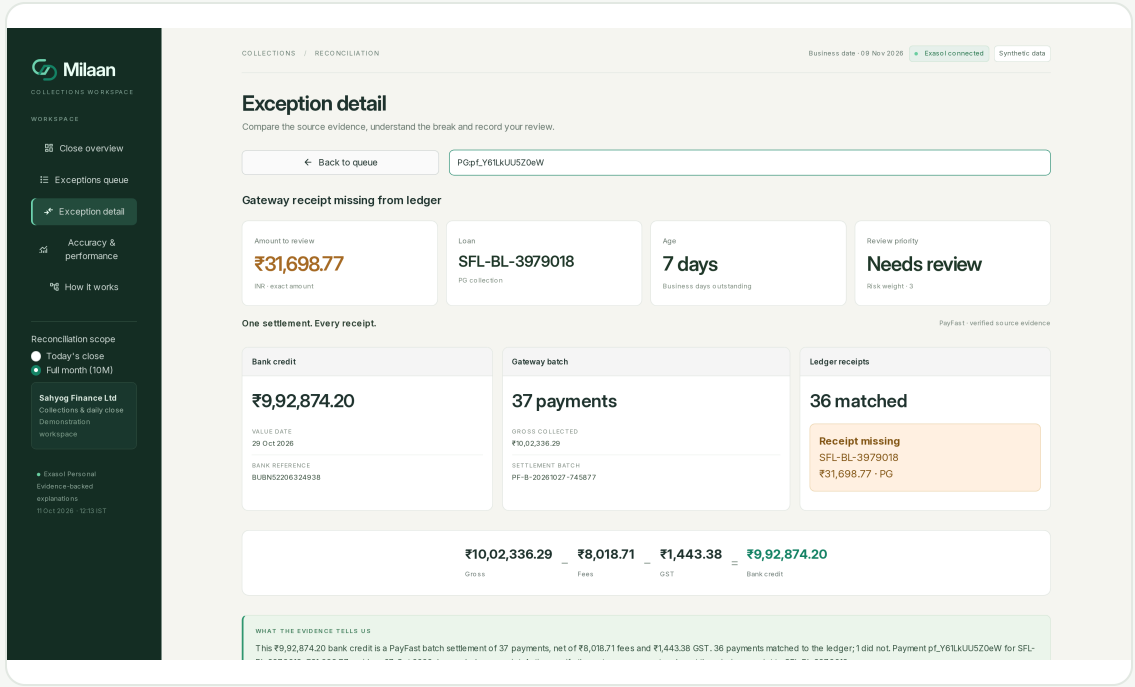
Bulk IMPORT · hash joins · REGEXP · windows · Python scalar · Lua SET UDFs

EXACT: 22.15s · PG_BATCH: 7.63s · FEE_AWARE: 9.15s · TOLERANT: 1.79s · FUZZY: 0.91s · SPLIT_MERGE: 3.11s · CLASSIFY: 9.88s

Database-side results, validation and scoring included. Explanations and profiling flush follow the timer.

Review the money that matters





Prioritise → inspect evidence → accept, reject or assign

Every review records the analyst, timestamp and reasoning in Exasol.

Accuracy, tested four ways

1M dataset	Precision	Recall	False links
Normal	100.000%	100.000%	0
Blind	100.000%	100.000%	0
Hard (tuned)	100.000%	85.872%	0
Hard (blind)	100.000%	85.867%	0

10M normal: 100.000% precision · 100.000% recall · 98.538% source membership

Normal and reused blind are regression checks. Hard (tuned) informed reversal handling; hard (blind) uses newly unseen seed 20261105.

Trust comes from evidence

Deterministic explanations
Amounts, dates and rules come from stored evidence. No external inference service.

Human review
The analyst accepts a recommendation, rejects it or assigns an owner.

Auditable decisions
Run-linked evidence and review history remain in Exasol.

A final membership check rejects conflicting match groups.

A practical path to impact

First: independently labelled validation with an NBFC partner.

Next: live feeds, real holiday calendars, authenticated review roles and approved ledger write-back.

Then: measure larger datasets and concurrent operations.

Similar reconciliation workflows exist in banks, insurers, fintechs and gateway-based merchants.

Milaan

Close with confidence

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https://github.com/Hemanth-Chilakala/milaan

Demo video: docs/video/demo.mp4

MIT · All data is synthetic

Appendix: limitations & methodology

- Seeded, controlled data; production accuracy needs independent partner labels.
- Reversal pairs excluded before matching; late debits reopen prior groups.
- Split search: at most 12 candidates, subsets of two or three.
- Single node and one worker; no real ledger connector or review authentication.
- Timing includes preparation, results, validation and scoring; browser/recording load changes it.
- Fixture and three consecutive 10M closes agree with month; full 30-day 10M replay unmeasured.
- Codespace rebuild and clean checkout tested; macOS/WSL remain unverified.

Run IDs, profiles and raw observations: docs/benchmarks.md